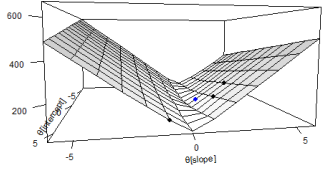
Introduction to Machine Learning

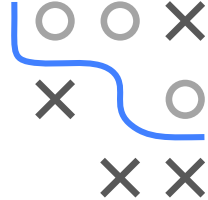
ML-Basics
Losses & Risk Minimization



Learning goals

- Know concept of loss function
- Understand relationship between loss and risk
- Understand relationship between risk minimization and finding best model

HOW TO EVALUATE MODELS



- When training a learner, we optimize over the hypothesis space, to find the function that fits the training data best.
- We are looking for a function, with which the predicted output per training point is as close as possible to the observed label.

Features x		Target y
People in Office (Feature 1) x_1	Salary (Feature 2) x_2	Worked Minutes Week (Target Variable)
4	4300 €	2220
12	2700 €	1800
5	3100 €	1920

$\mathcal{D}_{\text{train}}$

$\overset{?}{\approx}$

Prediction $\hat{y}$
Worked Minutes Week (Target Variable)
2588
1644
1870

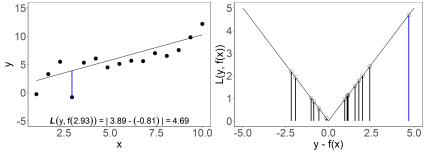
- Need to measure this pointnwise difference precisely.

LOSS

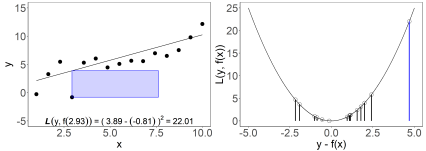
The **loss function** $L(y, f(\mathbf{x}))$ quantifies the "quality" of the prediction $f(\mathbf{x})$ of a single observation $\mathbf{x}$:

$$L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}.$$

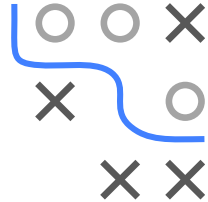
In regression, we can use the absolute loss $L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$;



or the L2-loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$:



RISK OF A MODEL



- The (theoretical) **risk** associated with a certain hypothesis f measured by a loss function $L(y, f(\mathbf{x}))$ is the **expected loss**

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}.$$

- This is the average error we incur when we use f on data from $\mathbb{P}_{xy}$.
- Goal in ML: Find a hypothesis $f \in \mathcal{H}$ that **minimizes** risk.

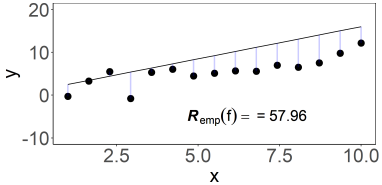
EMPIRICAL RISK

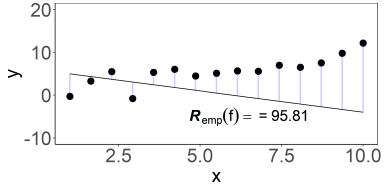
To evaluate, how well a given function f matches our training data, we now simply sum up all pointwise losses.

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

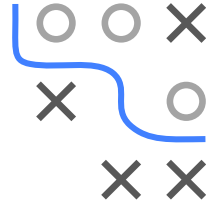
This **empirical risk function** allows us to associate one quality score with each $f \in \mathcal{H}$, which encodes how well it fits the training data.

$$\mathcal{R}_{\text{emp}} : \mathcal{H} \to \mathbb{R}$$





EMPIRICAL RISK MINIMIZATION

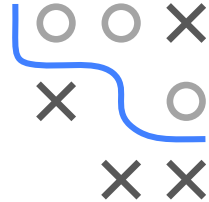


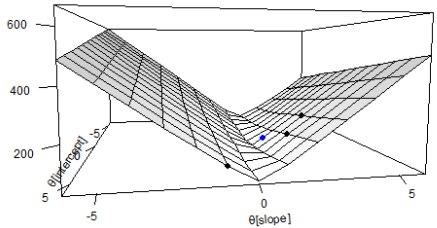
- The best model is the model with the smallest risk.
- For finite $\mathcal{H}$, we could tabulate all models and select the best.

Model	$\boldsymbol{\theta}_{intercept}$	$\boldsymbol{\theta}_{slope}$	$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
f_4	1	1.5	57.96

EMPIRICAL RISK MINIMIZATION

- But usually $\mathcal{H}$ is infinitely large.
- Instead we can consider the risk surface w.r.t. the parameters $\boldsymbol{\theta}$ (i.e., the visualization of $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$).



$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) : \mathbb{R}^d \to \mathbb{R}$$

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